& Relation b/w CSP and COP

* Minimise $2x+y$

such that $x+y \leq 4$

$x \neq y$

$x, y \in \{1, 2, 3\}$

} CO

CO is generalization of CSP

&

$A = \{1, 2, 3\}$

$B = \{1, 2, 3\}$

$C = \{1, 2, 3\}$

$A > B$

$B = C$

} Pure CSP problem

as there is no optimization function

$f(x) = 0$

Man $f(A, B, C) = A + B + C$ → Now its CO

$[3, 2, 2]$ → 10

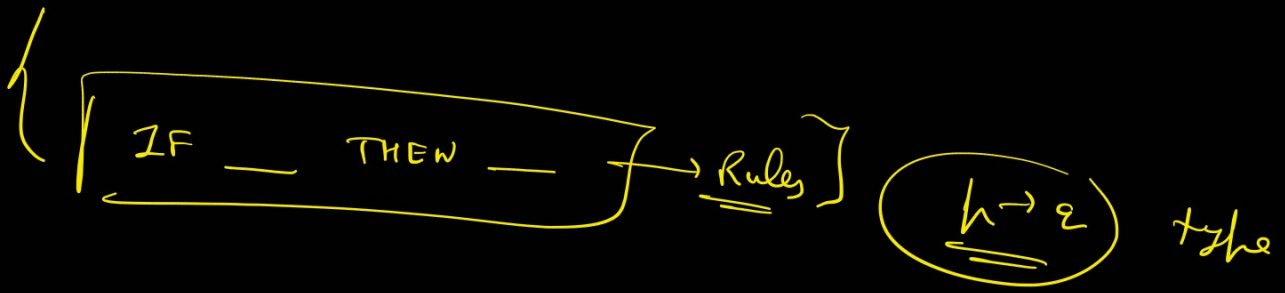
just adding an objective function converts it to CO

- Rule Based System [1960]
  - Forward-chaining (production system)
  - Backward chaining

Core math is predicate logic

in 1960, there was an Instant system based algo

called RETE ⟶ based on forward chaining



After Rule-based system came neural n/w

because there was a problem called backpropagation

Then there was another thing called fuzzy logic (1970)

Let's say

today is very hot

How to represent this in boolean logic.

Backward chaining in 1985 By Jeffy Hinton

the paper called Multilayer Perceptron.

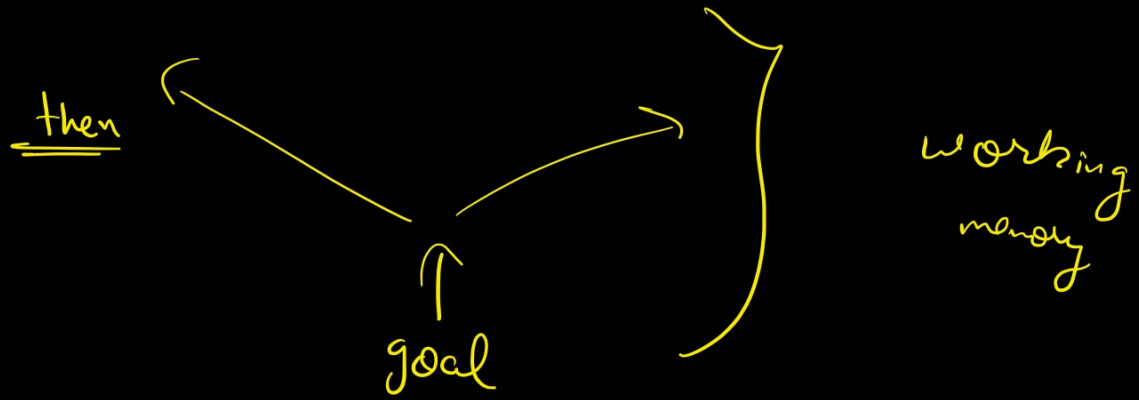
Forward Reasoning / Forward Chaining

A data-driven approach that starts with known facts and works forward to discover what conclusions can be drawn

Match current facts against rule conditions
↓
builds a conflict set
↓
applies conflict resolution to select one rule
↓
fires the rule

Backward Reasoning

↳ some goal is given, and we will check if the goal is true or not



formal defn ⟶ A goal driven approach that starts with specific goal and works backward to see if

existing facts support it.

Working Memory → a cache buffer stores the recently used facts.

rest of them stores in disk

Unification

Likes (John, x)

Likes (John, Mary)

Likes (John, Mary)

Human (x) ⟶ Mortal (x)

Human (Socrates)

Mortal (Socrates)

[Resolution]

Fact Bird (tweety) ⟶ tweety is a bird

Rule

Bird (x) ⟶ Flies (x)

Flies (Tweety) [Goal]

Check if this is true or not?

Sol Check By Modus Ponen & Modus Tollen